NETWORK MANAGEMENT 316D PRACTICAL ASSIGNMENT

Group 7

INTRODUCTION

This assignment focuses on the design and implementation of a secure and reliable network infrastructure for Company X, a growing organization with its head office located in Gauteng and additional branches planned for Cape Town, Free State, and Eastern Cape. As the newly appointed IT Managers, our responsibility was to design a complete network solution that supports communication, resource sharing, centralized management, and secure access across all branches.

Our network infrastructure design ensures efficient communication between departments and branch offices while improving productivity, security, scalability, and network performance. The project includes the planning of network topology, IP addressing and subnetting, server configuration, Active Directory implementation, organizational unit management, and remote desktop connectivity.

The practical component of the assignment focuses mainly on the Gauteng head office due to limited laboratory resources. The implementation includes configuring a domain controller, creating Active Directory organizational units for the IT, HR, and Procurement departments, applying Group Policies, managing user permissions, and testing network functionality. We explored the role of Artificial Intelligence (AI) in network management and the use of technology to promote indigenous South African languages through subtitled video presentations.

SECTION A

1. Network Topology Design

We have multiple Servers and the cloud for high availability, ensuring system can be available for all time in case one is affected .

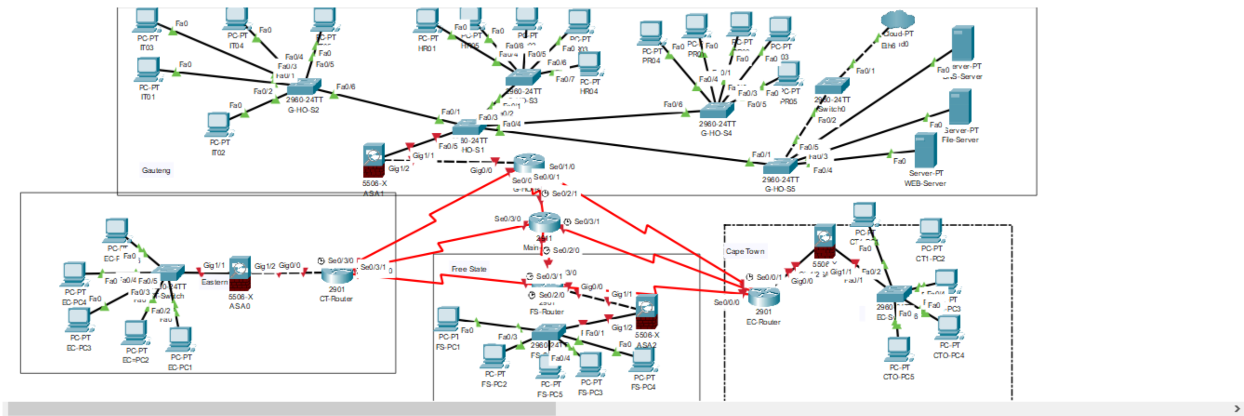
We have cloud because of incase there is natural disasters or the company is hacked, the database is always there as a backup.

For meeting the Policy of Service Level Agreements in the CompanyX .

We have multiple switches in case the others encounter a problem then taken for repair, it will not affect the running of CompanyX.

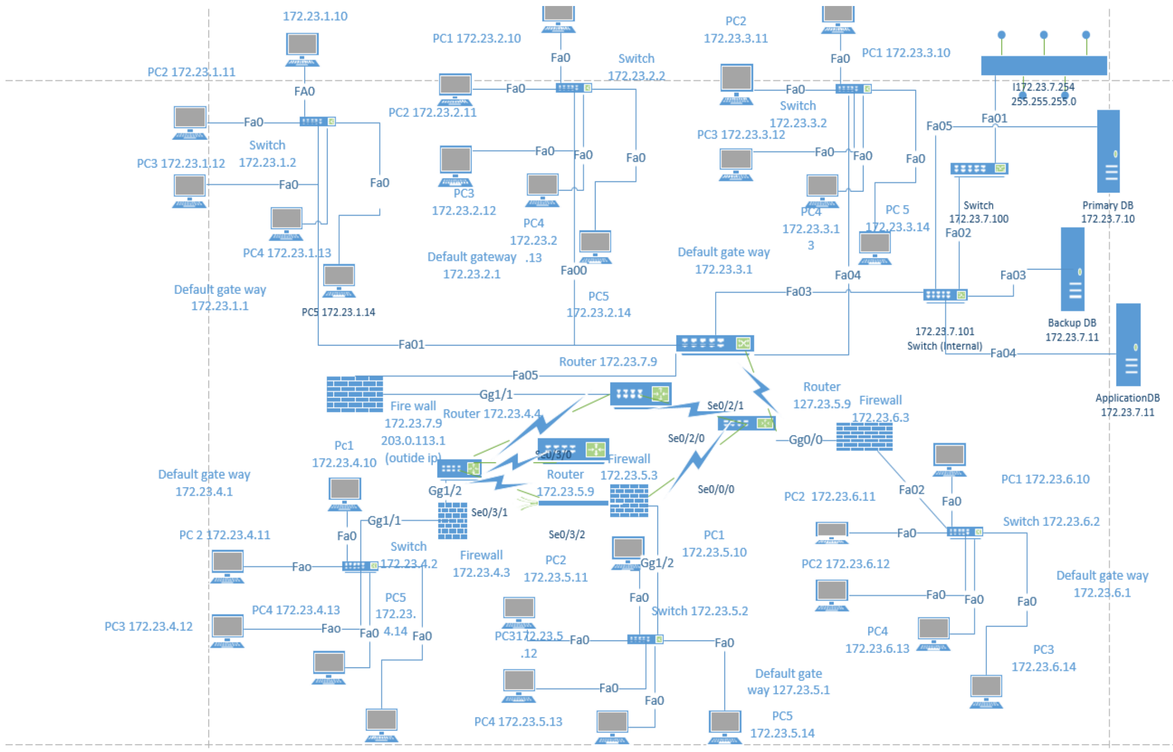
Firewall is a layer of security that ensures that the incoming and outgoing traffic is secure.

The multiple are interconnected routers are there, for redundancy so that if one router fails the services never stops.

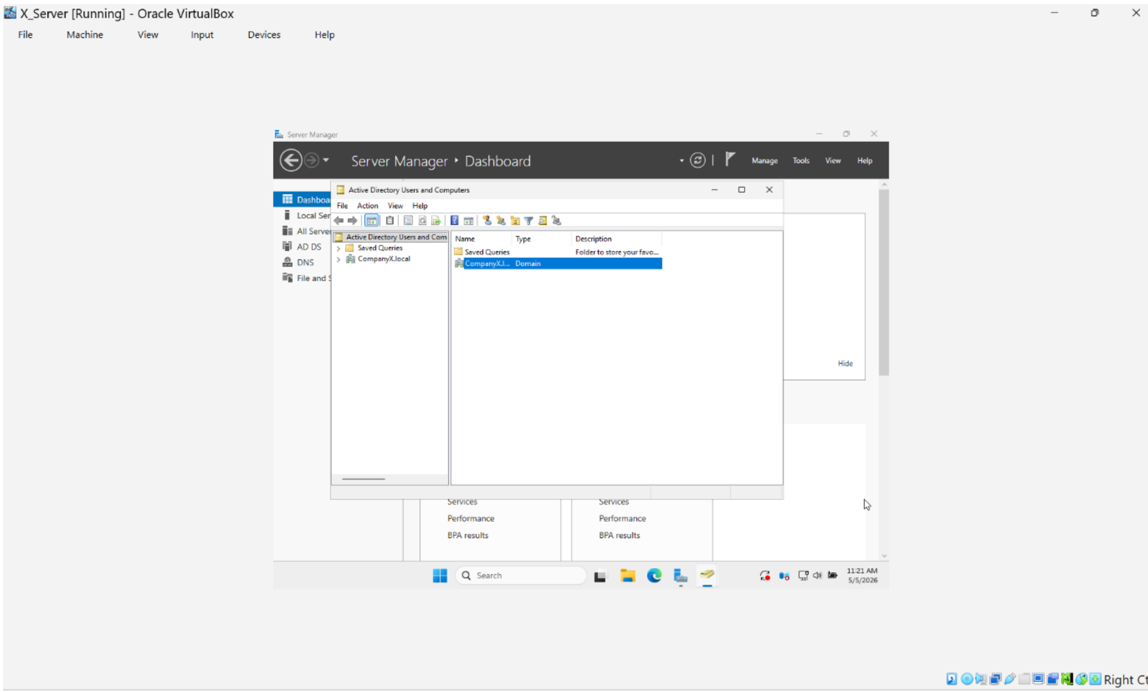


2. IP Addressing and Subnetting

We selected a unique IP range for Group 7 .Using 172.23.0.0/16 with multiple /24 subnets. Below it is the complete Ip configuration of the network.

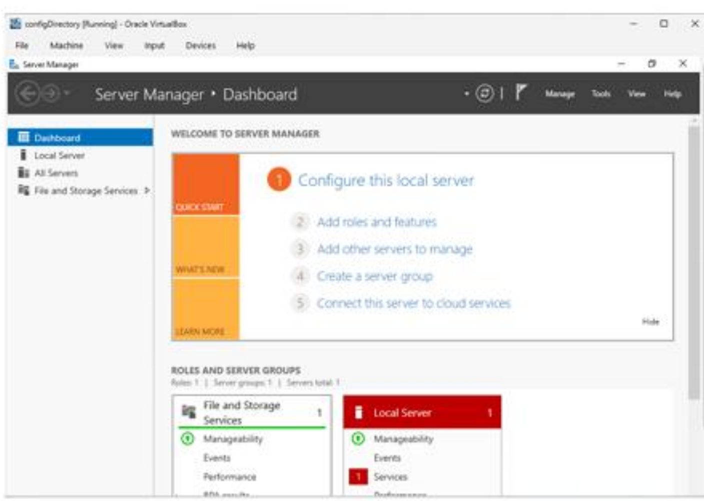


3.1Figure 3.1 – Domain Configuration (CompanyX.local)

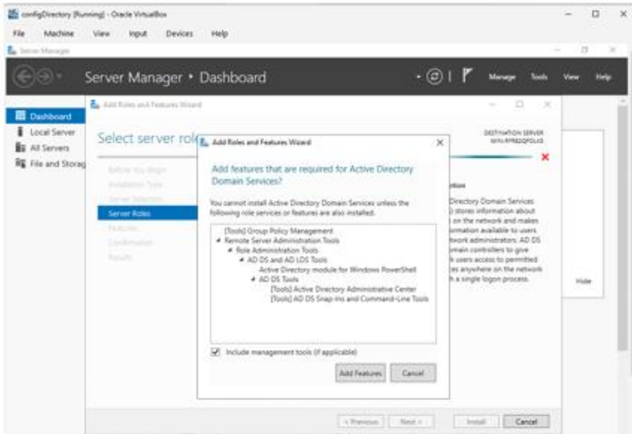


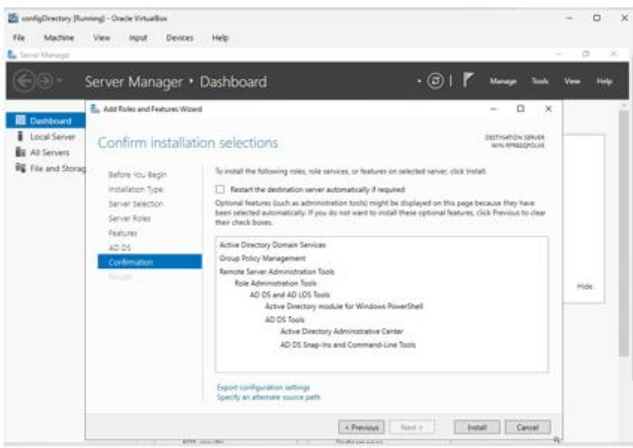
3.2. Creation and Configuration of an Active Directory for CompanyX at Head office

Step 1: Open service manager, under dashboard click "add roles and features"

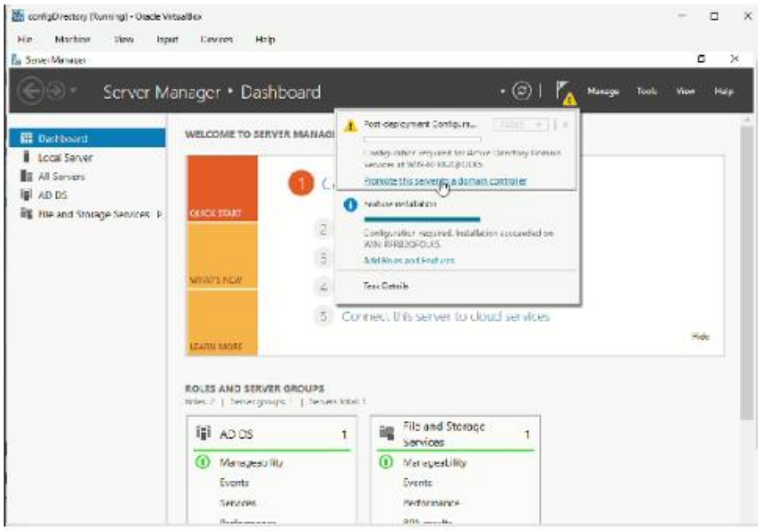


Step 2: Select “Active Directory Domain Services” then add all the features required for these services and click install.

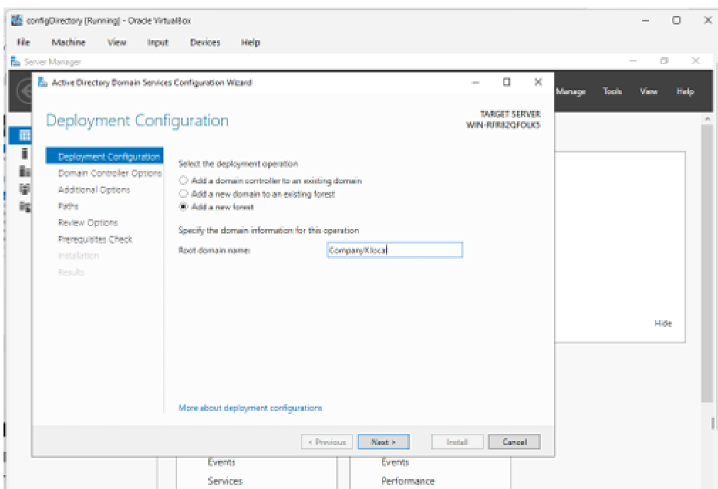


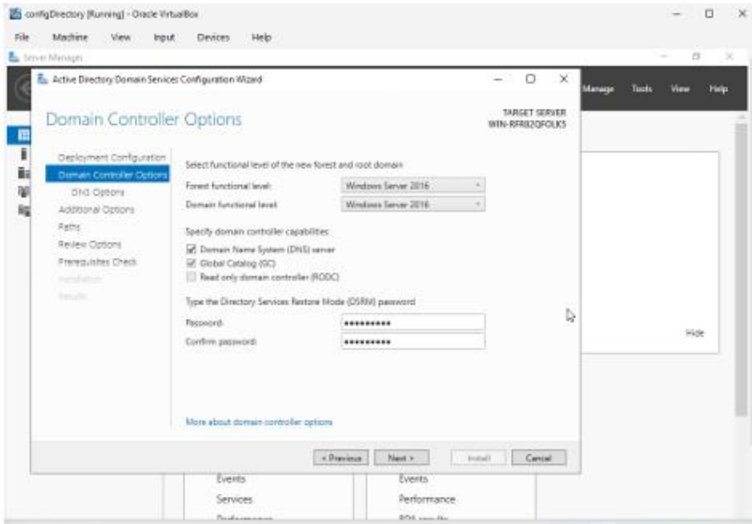


Step 3: After installation click “Promote this Server to a Domain Controller”

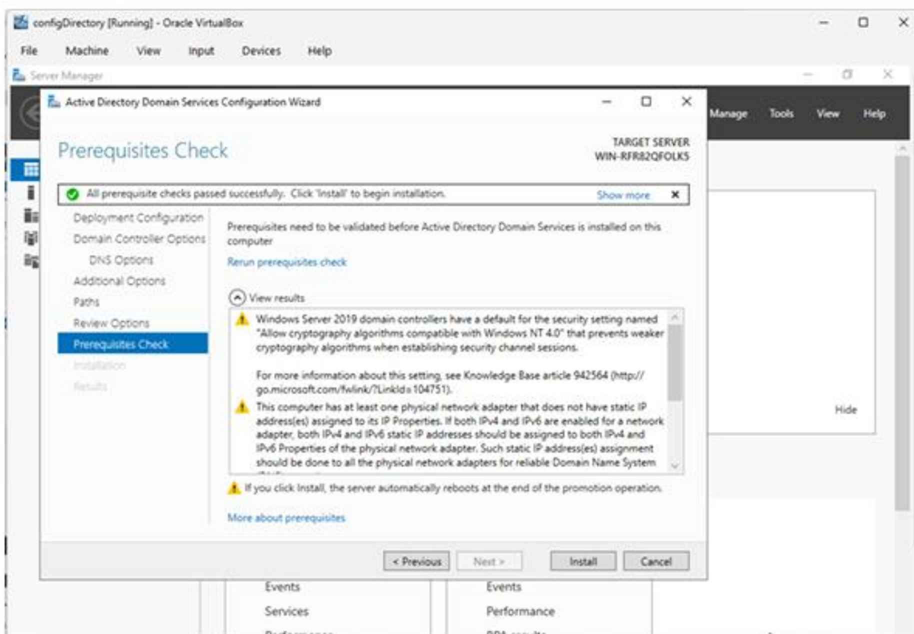


Step 4: Choose “Add a new forest” then assign the domain name and set Directory Services Restore Mode password.

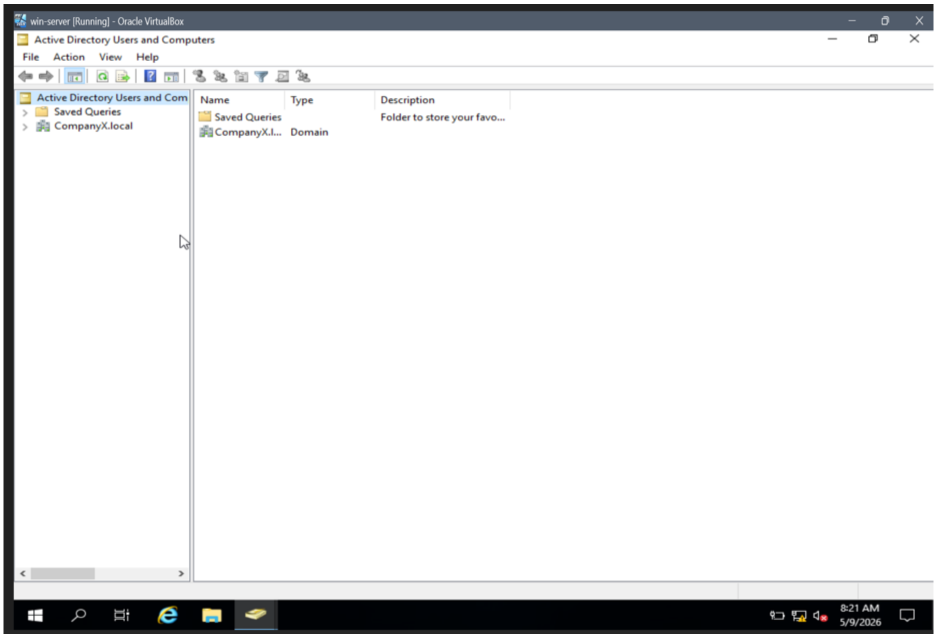




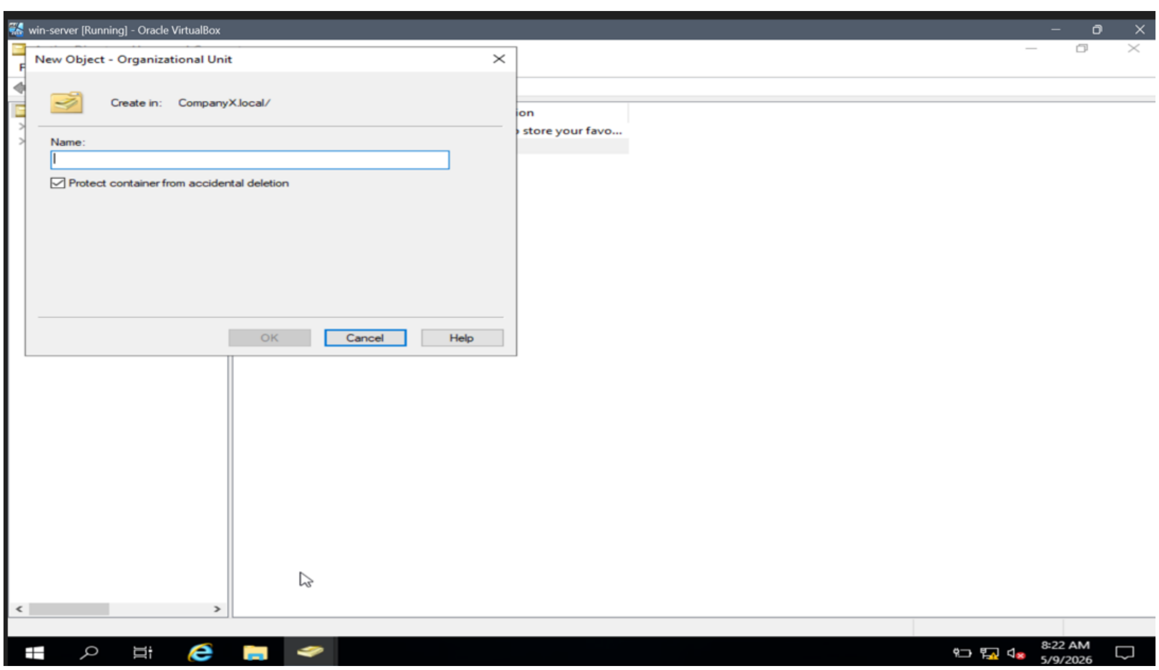
Step 5: Install and reboot to activate the directory.



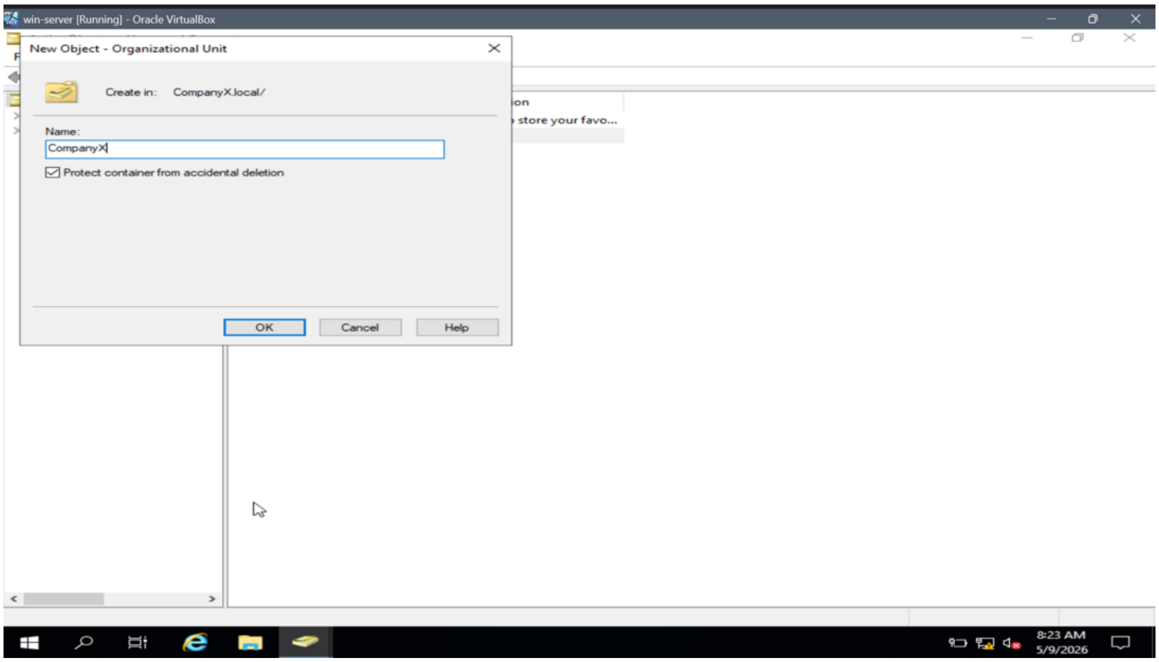
3.3 Active Directory Configuration

a. 

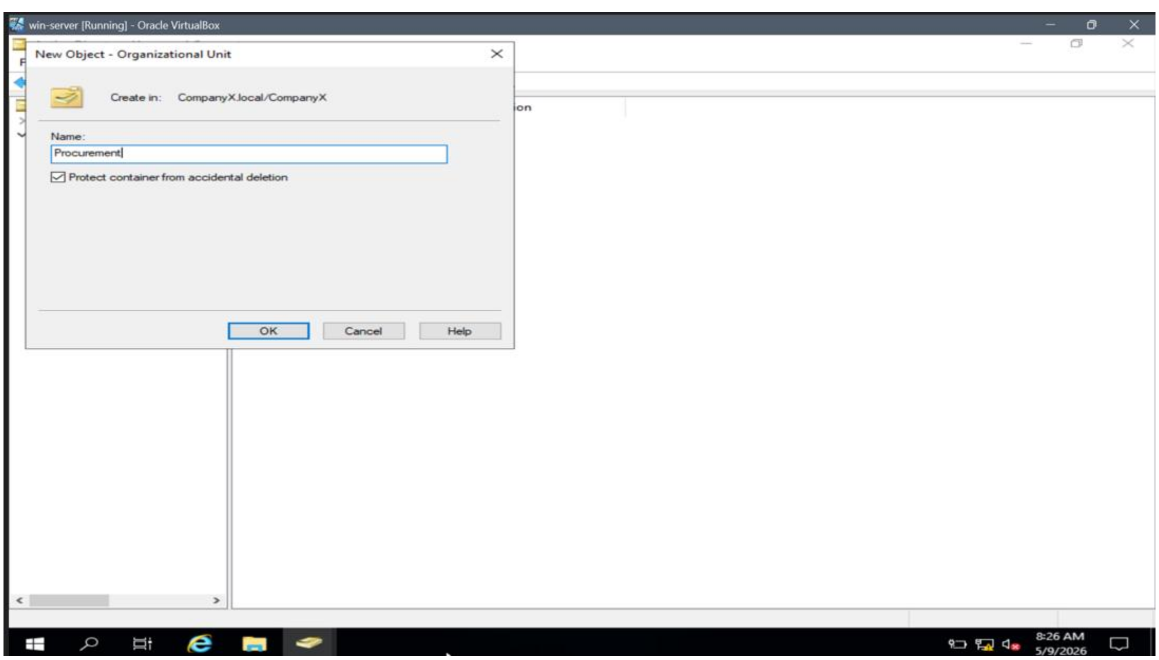
b.



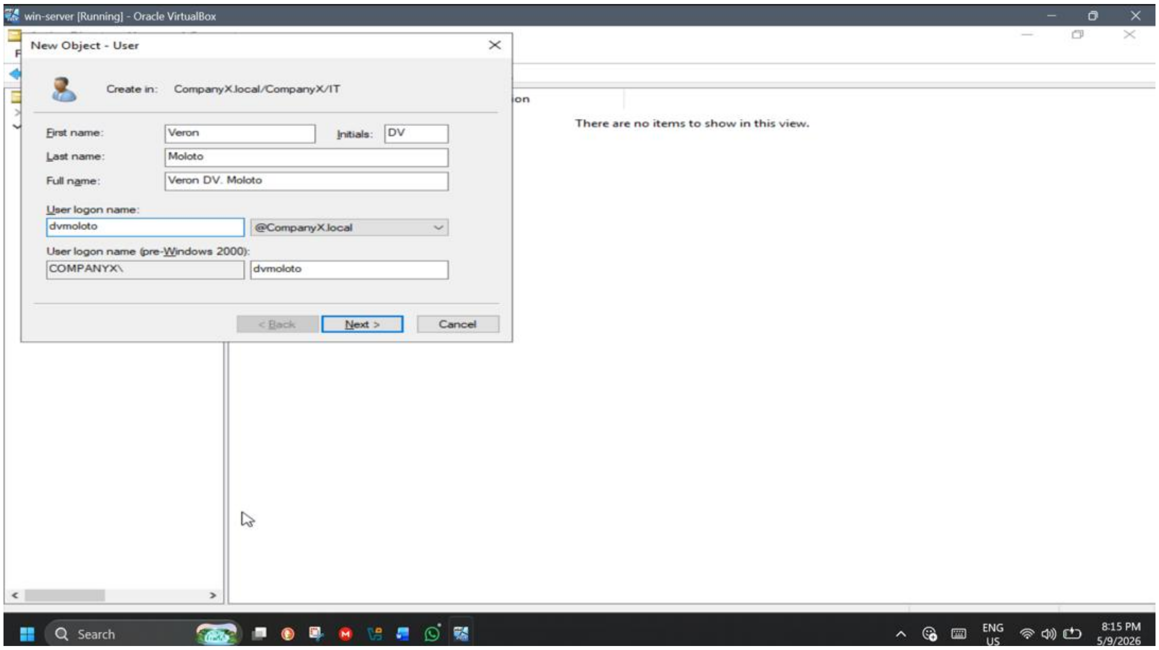
c.


win-server [Running] - Oracle VirtualBox
New Object - Organizational Unit
Create in: CompanyX.local/
Name:
CompanyX
Protect container from accidental deletion
OK
Cancel
Help
8:23 AM
5/9/2026

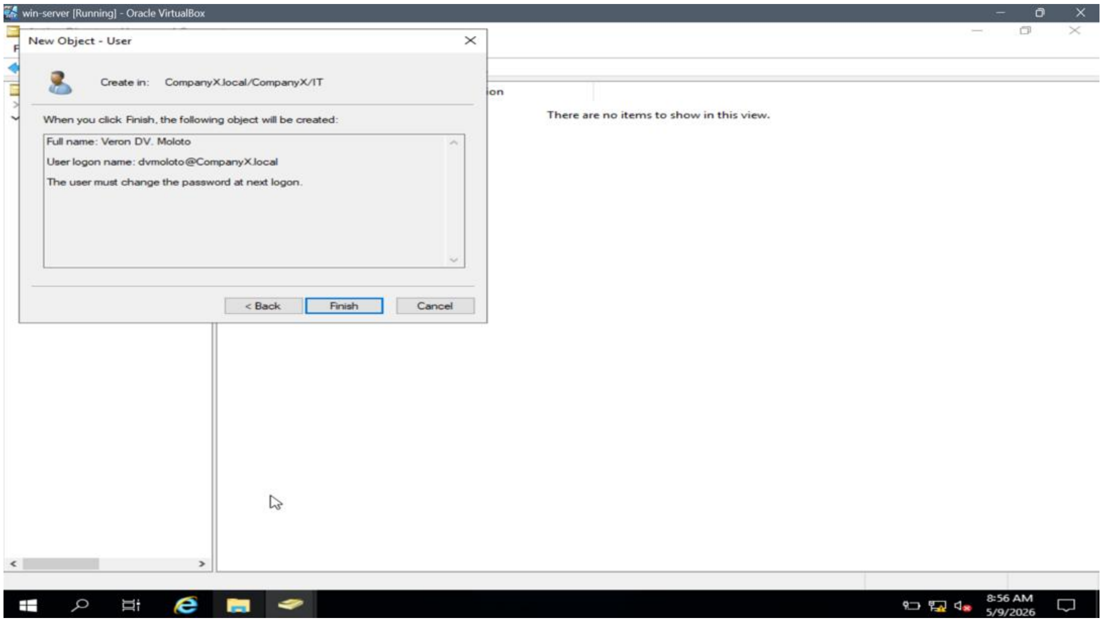
d.


win-server [Running] - Oracle VirtualBox
New Object - Organizational Unit
Create in: CompanyX.local/CompanyX
Name:
Procurement
Protect container from accidental deletion
OK
Cancel
Help
8:26 AM
5/9/2026

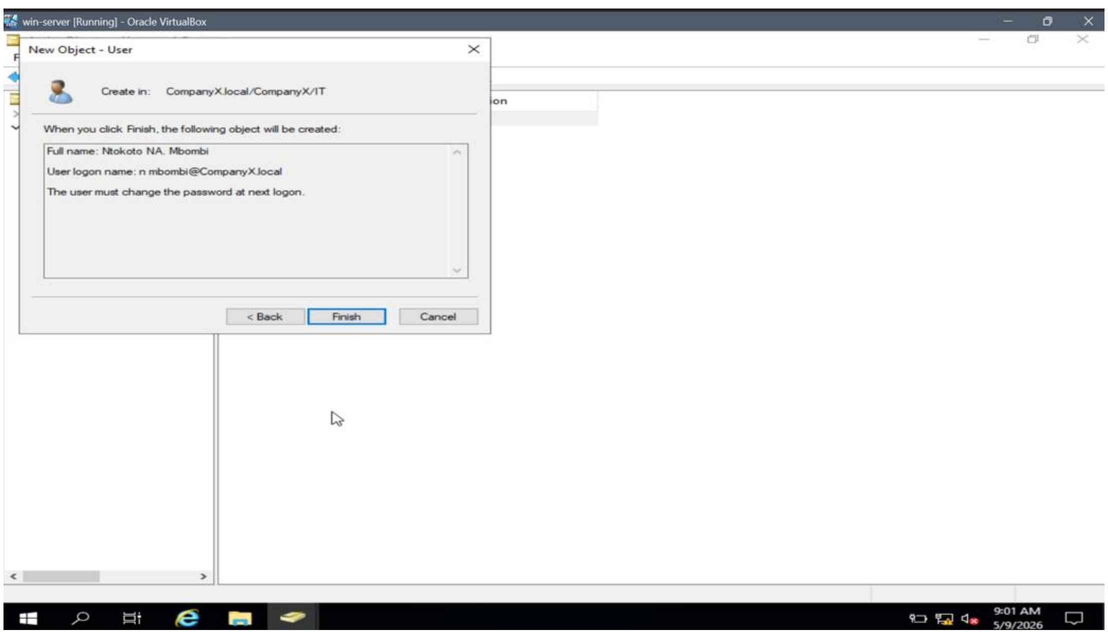
e.


win-server [Running] - Oracle VirtualBox
New Object - User
Create in: CompanyX.local/CompanyX/IT
First name: Veron
Initials: DV
Last name: Moloto
Full name: Veron DV. Moloto
User logon name:
dvmoloto
@CompanyX.local
User logon name (pre-Windows 2000):
COMPANYX\
dvmoloto
< Back
Next >
Cancel
There are no items to show in this view.
Search
ENG
US
8:15 PM
5/9/2026

f.


win-server [Running] - Oracle VirtualBox
New Object - User
Create in: CompanyX.local/CompanyX/IT
When you click Finish, the following object will be created:
Full name: Veron DV. Moloto
User logon name: dvmoloto@CompanyX.local
The user must change the password at next logon.
< Back
Finish
Cancel
There are no items to show in this view.
8:56 AM
5/9/2026

g.


win-server [Running] - Oracle VirtualBox
New Object - User
Create in: CompanyX.local/CompanyX/IT
When you click Finish, the following object will be created:
Full name: Ntokoto NA. Mbombi
User logon name: n mbombi@CompanyX.local
The user must change the password at next logon.
< Back
Finish
Cancel
9:01 AM
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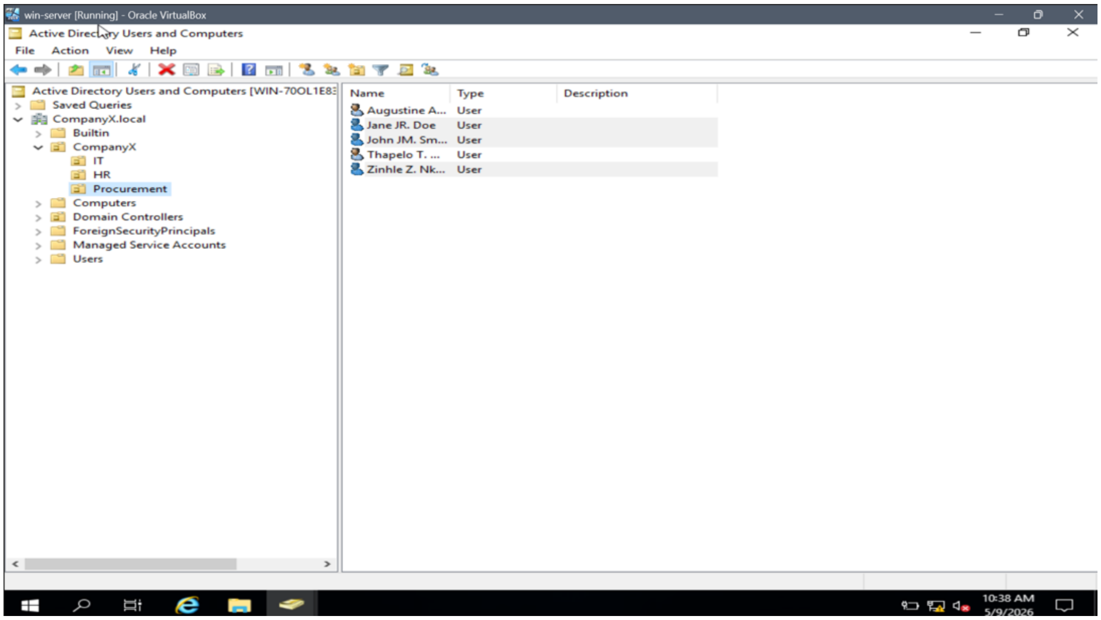
h.



win-server [Running] - Oracle VirtualBox
Active Directory Users and Computers
File Action View Help
Active Directory Users and Computers [WIN-70OL1E8
Saved Queries
CompanyX.local
Builtin
CompanyX
IT
HR
Procurement
Computers
Domain Controllers
ForeignSecurityPrincipals
Managed Service Accounts
Users
Name
Type
Description
Augustine A...
User
Jane JR. Doe
User
John JM. Sm...
User
Thapelo T. ...
User
Zinhle Z. Nk...
User
10:38 AM
5/9/2026

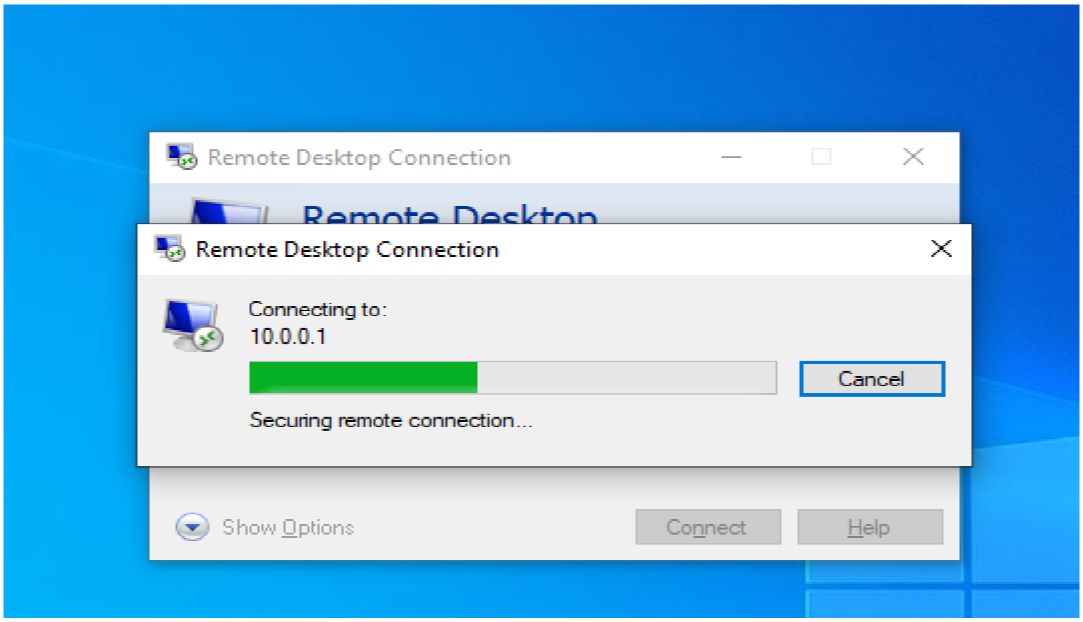
3.4. User Accounts and Group Policies

Show ipconfig on the Server to confirm its IP address, which is 10.0.0.1. This is the address that Client machines use to connect via Remote Desktop.

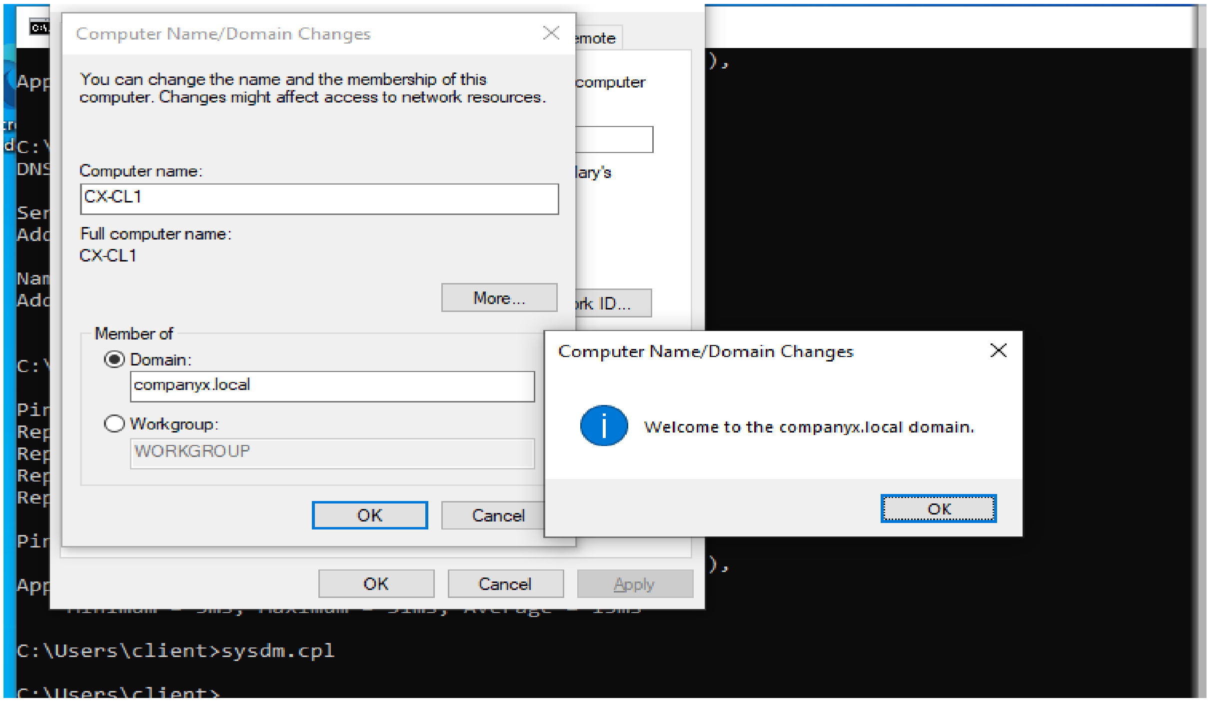
```
C:\Users\Administrator>ipconfig

Windows IP Configuration


Ethernet adapter Ethernet:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::efaa:b668:942a:5cfe%6
   IPv4 Address. . . . . . . . . . . : 10.0.0.1
   Subnet Mask . . . . . . . . . . . : 255.0.0.0
   Default Gateway . . . . . . . . . :

C:\Users\Administrator>
```

```
C:\Users\client>ping 10.0.0.1

Pinging 10.0.0.1 with 32 bytes of data:
Reply from 10.0.0.1: bytes=32 time=8ms TTL=128
Reply from 10.0.0.1: bytes=32 time=5ms TTL=128
Reply from 10.0.0.1: bytes=32 time=31ms TTL=128
Reply from 10.0.0.1: bytes=32 time=19ms TTL=128

Ping statistics for 10.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 5ms, Maximum = 31ms, Average = 15ms

C:\Users\client>
```

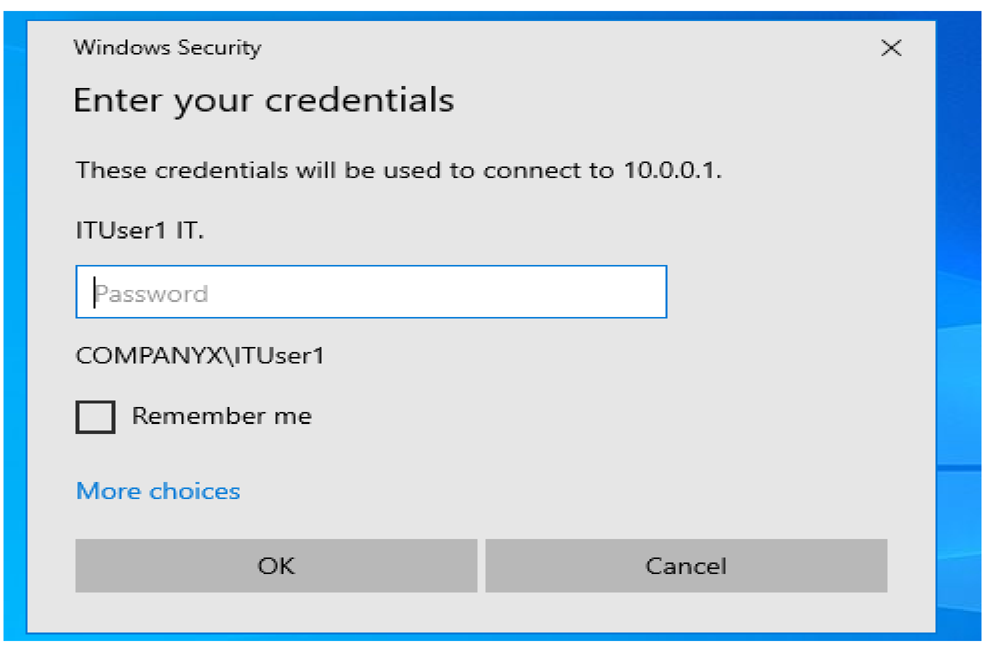
- The user on the Client machine has entered the Server's IP address (10.0.0.1) into the Remote Desktop Connection tool
- Client machine is connecting to the Server at IP 10.0.0.1, The Client PC can reach the Server across the network and the connection is secure



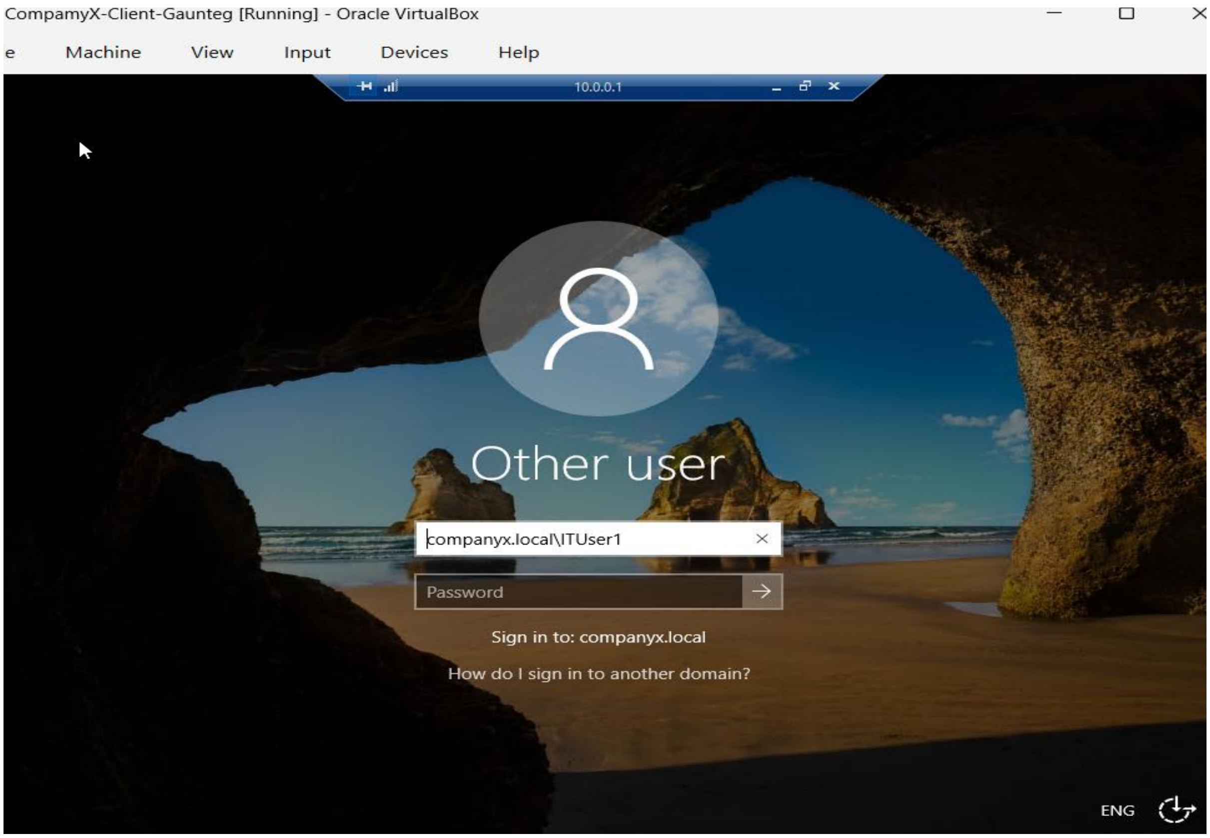
- Adding the Client PC to the domain so that Active Directory will manage it and AD users like IT personnel can use their accounts to access the client machine and implement group policies

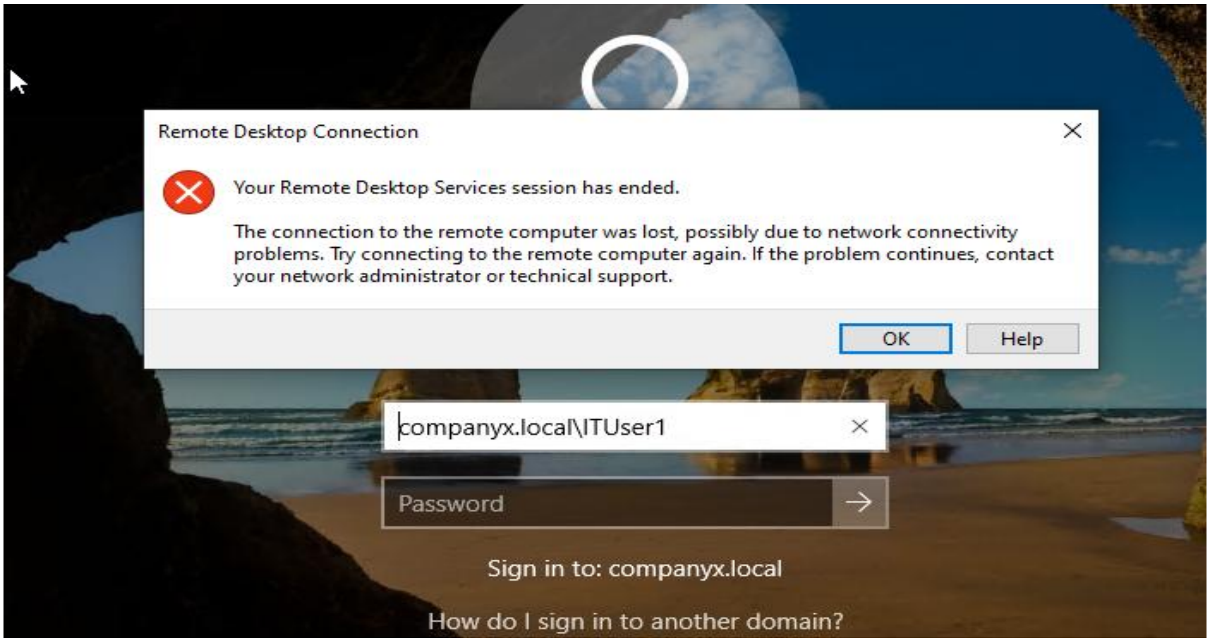


- Windows Security login prompt during a Remote Desktop connection from the Client PC to the Server (10.0.0.1).
- The Client machine is asking the User to enter their domain credentials, it ensures AD authentication, and only authorized users will be able to access the Server



- The log in screen shows "Other user" with username companyx.local\ITUser1. The domain is companyx.local, which indicates that the Client has successfully joined the domain

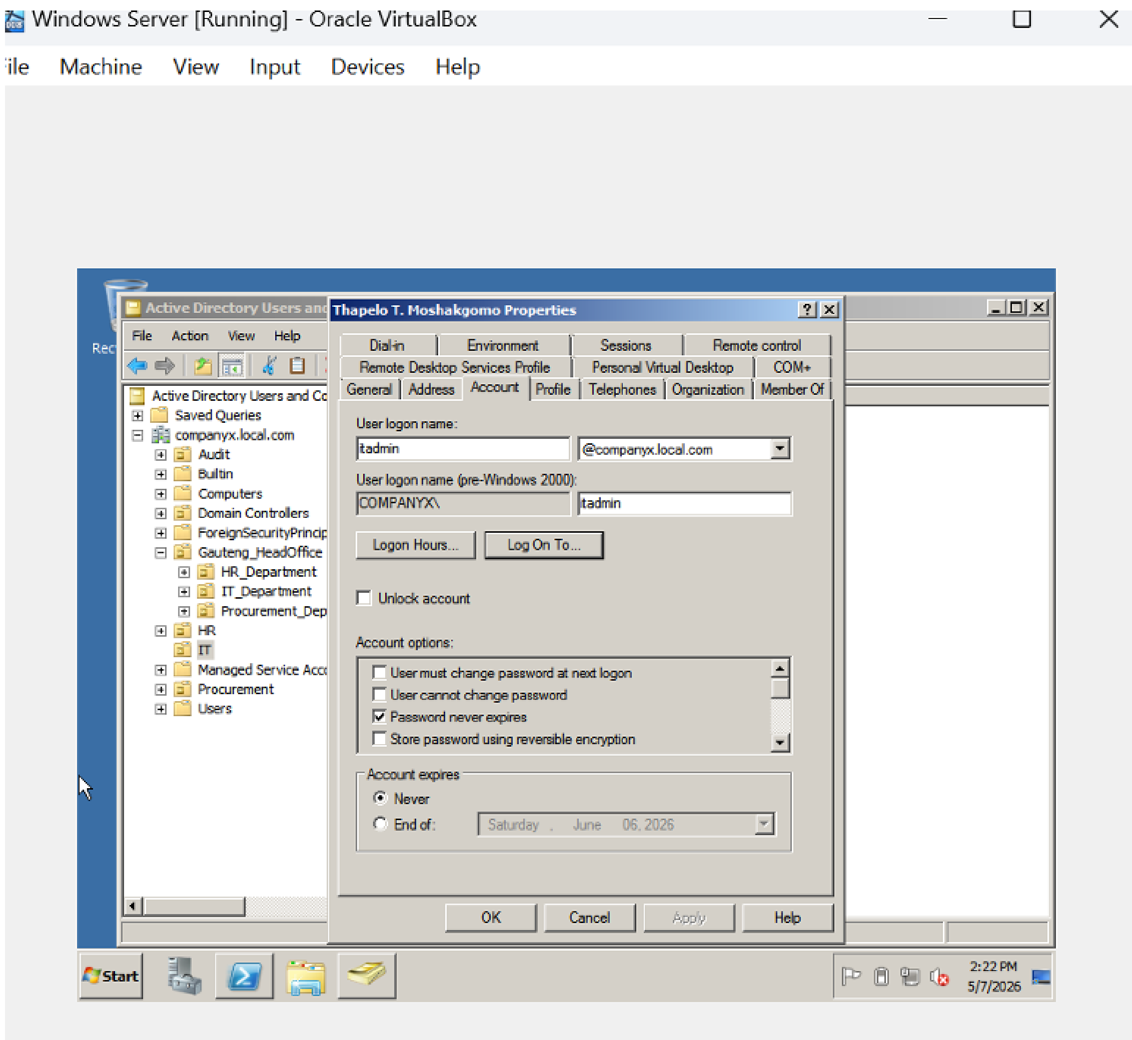



Remote Desktop Connection
Your Remote Desktop Services session has ended.
The connection to the remote computer was lost, possibly due to network connectivity problems. Try connecting to the remote computer again. If the problem continues, contact your network administrator or technical support.
OK
Help
companyx.local\ITUser1
Password
Sign in to: companyx.local
How do I sign in to another domain?

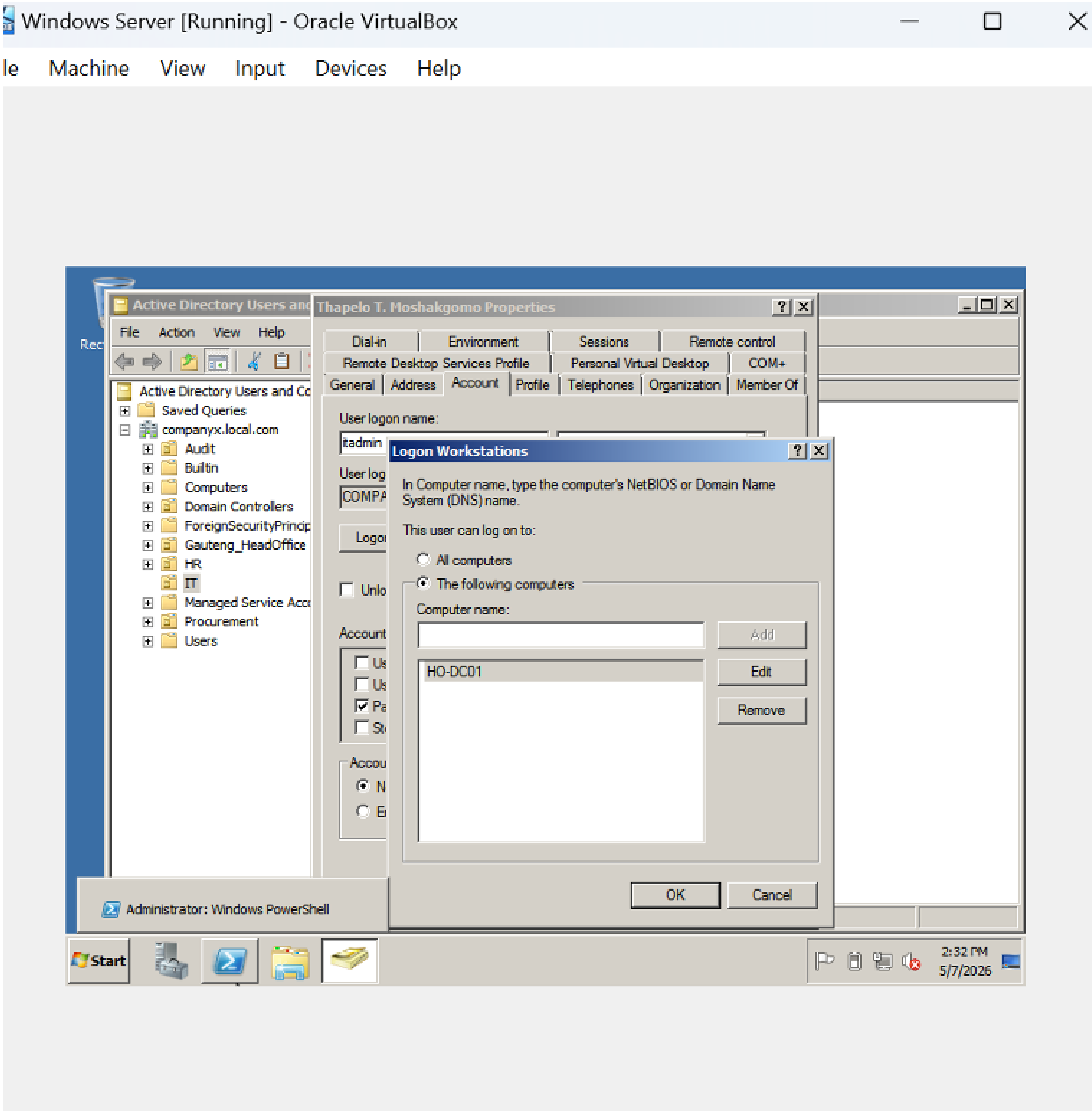
3.5. Remote Desktop Configuration

User access control was implemented through Account Tabs, then proceed to "Log On To" button allowing the users from IT department to have access to both clients and server machine.

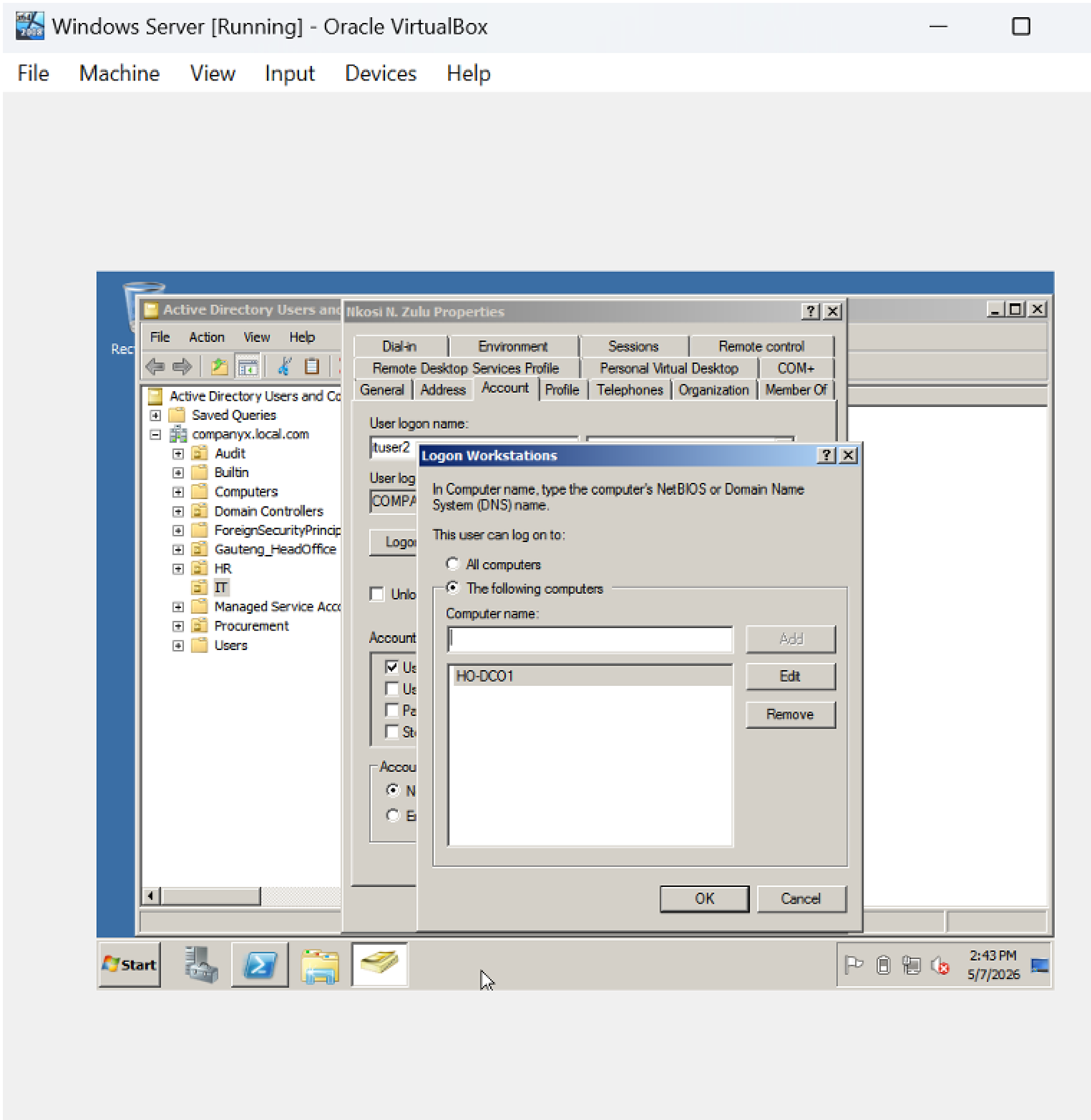
- **First user**



- **Second User**

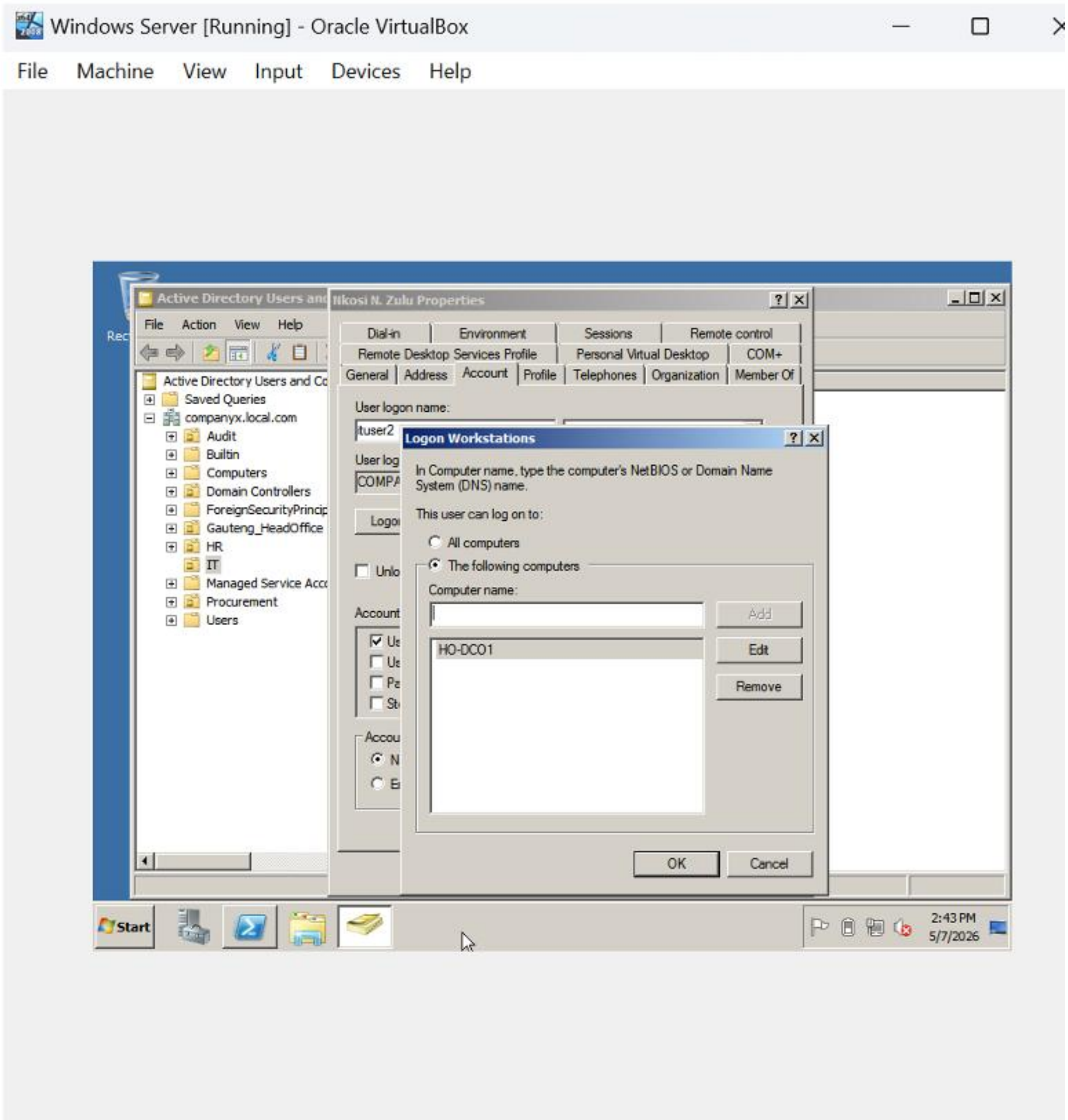

Windows Server [Running] - Oracle VirtualBox
le Machine View Input Devices Help
Active Directory Users and Computers
File Action View Help
Saved Queries
companyx.local.com
Audit
Builtin
Computers
Domain Controllers
ForeignSecurityPrincip
Gauteng_HeadOffice
HR
IT
Managed Service Acc
Procurement
Users
Thapelo T. Moshakgomo Properties
Dial-in
Environment
Sessions
Remote control
Remote Desktop Services Profile
Personal Virtual Desktop
COM+
General
Address
Account
Profile
Telephones
Organization
Member Of
User logon name:
tadmin
Logon Workstations
In Computer name, type the computer's NetBIOS or Domain Name System (DNS) name.
This user can log on to:
All computers
The following computers
Computer name:
Add
HO-DC01
Edit
Remove
OK
Cancel
Administrator: Windows PowerShell
Start
2:32 PM
5/7/2026

- **Third user**



Windows Server [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Active Directory Users and Co
Saved Queries
companyx.local.com
Audit
Builtin
Computers
Domain Controllers
ForeignSecurityPrincip
Gauteng_HeadOffice
HR
IT
Managed Service Acc
Procurement
Users
Nkosi N. Zulu Properties
Dial-in Environment Sessions Remote control
Remote Desktop Services Profile Personal Virtual Desktop COM+
General Address Account Profile Telephones Organization Member Of
User logon name:
ituser2
Logon Workstations
In Computer name, type the computer's NetBIOS or Domain Name System (DNS) name.
This user can log on to:
All computers
The following computers
Computer name:
Add
HO-DCO1
Edit
Remove
OK
Cancel
Start
2:43 PM
5/7/2026

- **Fourth User**



Windows Server [Running] - Oracle VirtualBox
File
Machine
View
Input
Devices
Help
Active Directory Users and
Nkosi N. Zulu Properties
Dial-in
Environment
Sessions
Remote control
Remote Desktop Services Profile
Personal Virtual Desktop
COM+
General
Address
Account
Profile
Telephones
Organization
Member Of
User logon name:
tuser2
Logon Workstations
In Computer name, type the computer's NetBIOS or Domain Name System (DNS) name.
This user can log on to:
All computers
The following computers
Computer name:
Add
HO-DC01
Edit
Remove
OK
Cancel
Active Directory Users and Co
Saved Queries
companyx.local.com
Audit
Builtin
Computers
Domain Controllers
ForeignSecurityPrincip
Gauteng_HeadOffice
HR
IT
Managed Service Acc
Procurement
Users
Start
2:43 PM
5/7/2026

4. References

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SECTION B

2.1.

LN Motubatse

230481580

AI Tools for Network Management

LN MOTUBATSE posted 12 May 2026 7:19 PM Subscribed

1. AI tool I recommend: CISCO AI CANVAS
2. Two reasons why it is suitable for network management:

- It dynamically generates insights, visualizations, and recommended actions based on Cisco's Deep Network Model.
- Enables distributed teams to work collaboratively within a single interface

3. Challenge or Limitation:

- Limited integrations for non-Cisco environments require additional configuration

less

0 Unread 0 Replies 0 Views

Replied to BM MPHETHI

LN MOTUBATSE

just now

Your discussion clearly explains how Cisco AI Network Analytics can support network management in different types of organizations. The point about proactive monitoring and predictive analytics shows how AI tools can help administrators identify network problems before they cause downtime.

The challenge you mentioned about cost and complexity also highlights an important issue organizations face when adopting AI technologies. Smaller organizations may struggle financially, while larger organizations may need time and expertise to integrate AI tools with existing systems.

Reply

Replied to PL MASHIANE

LN MOTUBATSE

just now

AIOps is presented clearly as an effective approach for improving network management in organizations of different sizes. Continuous monitoring and predictive fault detection help reduce downtime and allow faster responses to network problems before they become serious.

Automating tasks such as troubleshooting, device configuration, and performance monitoring can also improve efficiency and reduce the workload of IT staff. Implementation costs, infrastructure upgrades, and the need for skilled personnel remain important challenges that organizations must consider when adopting AI-based systems.

Reply

TP Mokapu

224602545

Juniper Mist AI

TP MOKAPU posted 11 May 2026 7:43 PM Subscribed

1. **AI Tool/Approach I Recommend:** Juniper Mist AI
2. **Reasons it is suitable:Automated Network Insights:** Mist AI uses machine learning to continuously monitor network performance and automatically detect anomalies, reducing manual troubleshooting. It provides real-time visibility into end-user connectivity and application performance, ensuring both large and small organizations can maintain reliable service.
3. **Challenge/limitation:** Organizations with legacy infrastructure may face challenges integrating Mist AI with existing systems, requiring additional investment in compatible hardware or training.

less

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TP MOKAPU

just now

Thank you for sharing your insights, SL Mbatha. I found your explanation clear and well-structured. I especially appreciate how you highlighted the practical benefits of the tool in improving network management efficiency. One point I'd like to add is that while the tool offers strong automation, organizations should also consider the training and adaptation required for staff to fully leverage its capabilities. Overall, your post gave me a better perspective on how AI can be applied in real-world scenarios.

Reply

TP MOKAPU

just now

You raise an important perspective, K. Tsotetsi. AI tools are indeed reshaping network management by enabling predictive monitoring, automated troubleshooting, and adaptive security responses. One challenge, however, is ensuring that these tools integrate smoothly with existing infrastructure and that administrators maintain oversight to avoid over-reliance on automation. It would be interesting to explore how these tools balance efficiency with accountability in real-world deployments.

Reply

T Moshakgomo

220317706

Network Management

T MOSHAKGOMO posted 12 May 2026 7:57 PM Subscribed

1. The AI-based tool I recommend is Cisco DNA Center, which uses Artificial Intelligence for IT Operations to manage and monitor networks efficiently.
2. Automated Network Monitoring and Troubleshooting- By continuously monitoring network devices and traffic, the tool proactively detects and resolves faults before they impact users, reducing downtime and increasing productivity. Scalability - The tool is suitable for both small and large organizations because it can manage many devices across multiple branches from a centralized dashboard
3. The high cost of deployment and maintenance poses a significant barrier to adopting AI-based network management tools, particularly for small organizations that may lack the budget for software licenses, infrastructure upgrades, and skilled IT staff. Additionally, these AI systems are only as reliable as the data they receive, inaccurate data can result in poor decision-making and false alarms

Reply to Thread

Filter by: Status: Unread | Clear filters

M MTSWENI

3 minutes ago

Cisco DNA Center is a powerful AI-driven network management tool. It offers clear benefits like **automated monitoring, proactive troubleshooting, and scalability across multiple branches**, which can greatly improve uptime and productivity.

However, the **high cost of deployment and maintenance** makes it challenging for smaller organizations, especially when factoring in licensing, infrastructure, and skilled staff. Also, its effectiveness depends heavily on **accurate data input**—poor data quality can lead to false alarms or wrong decisions.

Reply

224631600

Mbatha S.L

Discussions List › View Topic › View Thread

Cisco DNA Center with AI-driven analytics

SL MBATHA posted 09 May 2026 10:30 PM Subscribed

1. Cisco DNA Center with AI-driven analytics
2. **Automation & Efficiency**

   - AI automates repetitive tasks like device configuration, fault detection, and performance monitoring.
   - AI automates repetitive tasks like device configuration, fault detection, and performance monitoring.

   Predictive Insights

   - AI models analyze traffic patterns to predict potential failures or congestion before they happen.
   - Helps maintain uptime and ensures smooth service delivery for end-users.
3. **High Cost & Complexity**

   - Deploying Cisco DNA Center or similar AI-driven platforms requires significant investment in hardware, licensing, and skilled staff
   - Smaller organizations may struggle with affordability and expertise, making adoption slower compared to large enterprises.

Reply to Thread

Filter by: All Posts | Clear filters

SL MBATHA

Sat at 10:50 PM

Nice points raised in your discussion. I concur with your selection of Cisco AI Network Analytics with AIOps as one of the best options available. It combines the two aspects of automation and predictive monitoring and can be useful for both large corporations and small organizations.

Your first limitation on the ability to detect any suspicious activity in a network, traffic, and other network devices in real time is well-founded. Real-time detection is a unique characteristic that separates AI from other monitoring technologies. Similarly, your second advantage on the effectiveness and efficiency of such tools in automating processes cannot be overlooked. It minimizes human errors and conserves time.

Reply

SL MBATHA

Sat at 10:54 PM

I am convinced that you made a strong case for Cisco DNA Center. I really liked the way you highlighted its features in relation to automation. Device configuration, updates, and policies are all those processes that can be easily automated using an artificial intelligence system.

The importance of the ability to monitor the operation of a network in real-time and predict failures is another valid argument you provided. This is especially true when we talk about large-scale operations since early detection of problems helps save time and reputation.

However, I agree that there may be difficulties in implementing the technology into practice in small companies because of such factors as reliance on Cisco equipment and license requirements as well as the requirement for specialized personnel.

Reply

223719481

S MOLABA

AI Tools for Network Management

S MOLABA posted 12 May 2026 8:59 PM Subscribed

1. The AI-based approach I recommend for effective network management is Artificial Intelligence for IT Operations (AIOps) ;
2. It helps detect network problems automatically by using machine learning and intelligent monitoring. This helps network administrators identify faults quickly and reduce downtime. Alongside it improves efficiency through automation. It can automate network monitoring, traffic analysis, and troubleshooting, which saves time and reduces human errors in both small and large organizations
3. One major challenge of implementing AIOps is the high cost of deployment and maintenance. Many organizations may not have enough budget, infrastructure, or skilled personnel to properly implement and manage AI-based systems. In addition, employees may require training to understand and operate the AI tools effectively.

Reply to Thread

S MOLABA

just now

Agreed, PRTG's strength is the centralized view and flexibility to scale with unlimited servers. For large networks, the wide protocol support plus SMS/push notifications really help shift from reactive to proactive monitoring. I also like that the sensor-based licensing makes costs transparent. The downside is that at a certain scale, managing sensors and the infrastructure to run them becomes a project in itself.

Reply

S MOLABA

just now

You've illustrated very well. These platforms aggregate massive amounts of structured and unstructured data from multiple IT sources such as logs, metrics, alerts, monitoring tools, and tickets; advantages could be Faster detection and root cause analysis result in minimal downtime, Reduces manual monitoring and repetitive tasks, freeing teams for higher-value work "efficiency"....

Reply

Mello MA

231147900

Section A 1.2

AM MELLO posted 12 May 2026 9:49 PM Subscribed

1. I recommend Cisco AI Network Analystics for large organizations, and Domotz for small organizations. Cisco AI Network Analystics ,it is designed for enterprise environments with hundreds or thousands of devices. Domotz scales perfectly for 10 to 200 devices.

2. Proactive issue resolution : AI-based anomaly detection continuously learns normal network behavior , like traffic patterns, device response times, bandwidth usage.

Automated Root Cause Analysis :In large organizations, a single symptom can have dozens of potential causes. AI tools can ingest logs, SNMP traps, and flow data to correlate events and pinpoint whether the issue is a misconfigured switch, broadcast storm, or bandwidth saturation.

3. The problem might be data quality and infrastructure readiness . Many small organizations have old switches that don't support modern telemetry. The AI will give false alarms or miss real failures until you clean up your data and standardize SNMP. For a small budget, that upfront work is tough.

AM MELLO

just now

Solid choice with Cisco DNA Center. I really like how you highlighted AI-driven anomaly prediction and device fingerprinting those are game changers for stopping issues before users complain.

Your limitation about complexity and legacy device upgrades is on point.

Reply

AM MELLO

just now

I really like your AIOps recommendation , it's such a practical fit for both small and large organizations . The way you connected automated fault detection to reducing downtime is exactly why network managers are moving this direction.

Your challenge about high deployment cost and retraining staff is painfully real.

Reply

21955079

Mmapheto MA

NETWORK MANAGEMENT 316R (2026/1) MM

Print AI Tool for Managing Networks

NETWORK MANAGEMENT 316R (2026/1) Topic: AI Tools for Network Management Topic: AI Tools for Network Management

AI Tool for Managing Networks

Created by SL NDZUKULA on 12 May 2026 10:04 PM

1. Juniper Mist AI is the AI tool I have chosen for managing both large and small organization networks effectively

2. I have chosen Juniper Mist AI because:
   Cloud-Native Scalability - Juniper is built on a cloud native architecture, meaning it can scale easily across small branch offices or large enterprise networks and also organizations can manage WI-FI, WAN and LAN from a single AI-driven dashboard.
   Self-healing & Predictive Analytics - Juniper uses machine learning to automatically detect anomalies, predict failures, and even resolve issues before they impact users, this reduces downtime and improves reliability.

3. Challenge or Limitation of Juniper Mist AI:
   Data Privacy and Vendor Lock-In - Juniper Mist AI's cloud-centric architecture requires organizations to rely on Juniper for handling sensitive network telemetry, which may raise data privacy concerns. Furthermore, adopting the platform can lead to vendor lock-in, as migrating to alternative solutions later may involve significant cost, complexity, and reduced operational flexibility.

MA MMAPHETO
Posted 12 May 2026 10:14 PM

cloud-native scalability because it allows both small and large organizations to manage their networks efficiently from a centralized dashboard

224161670 NA Mbombi

AI tools for network management

NA MBOMBI posted 12 May 2026 11:27 PM Subscribed

One AI-based tool I recommend for effective network management is Cisco AI Network Analytics.

This tool is suitable for network management because:

1. It can automatically detect network problems before they become serious.
2. It improves network performance through intelligent monitoring and predictive analysis.

One challenge when implementing this tool in a real organization is the high cost of deployment and training staff to use the system effectively.

less

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Last post **11 minutes ago** by NM MTETWA

Opinion based on AI-based tool

NM MTETWA posted 12 May 2026 11:03 PM Subscribe

1. My recommended AI-based tool for effective network management is ***Cisco AI Network Analytics.***
2. The reasons why:
   - It monitors, manages and optimizes the network infrastructure in real time
   - It allows network administrators to make data-driven decisions about infrastructure upgrades, QoS policies, and load balancing.
3. Its challenge is that it has high implementation cost and infrastructure dependency. Deployment requires skilled Cisco-certified network engineers to configure assurance policies and interpret AI-generated insights correctly.

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NA MBOMBI

1 hour ago

I agree with your point that AI helps reduce network downtime. Your explanation about automation improving efficiency was very clear and relevant to modern networking technology.

Reply

resolutions , reducing mean time to repair significantly

3. Challenge or Limitation

-Vendor lock-in and integration issues

Cisco AI Assistant works best within a Cisco ecosystem. Organizations with multi-vendor infrastructures may face challenges integrating it seamlessly, which can reduce flexibility and increase costs

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NA MBOMBI

1 hour ago

You made a good point about AI improving network security. I also agree that real-time monitoring helps network administrators respond faster to threats and failures.

Conclusion

- In conclusion, this assignment successfully demonstrated the design and implementation of a secure and reliable network infrastructure for CompanyX. The project included the development of a complete network topology connecting the head office in Gauteng with branch offices located in Cape Town, Free State, and Eastern Cape. Through subnetting and proper IP addressing, the network was organized efficiently to support communication between all departments and branches.
- The practical implementation at the Gauteng head office successfully achieved the installation and configuration of a domain controller, Active Directory services, organizational units, user accounts, Group Policies, and Remote Desktop connectivity. Security and access control measures were also implemented to ensure that only authorized users could access specific systems and resources. In addition, dedicated home drives and desktop restrictions were configured according to departmental requirements.
- Furthermore, the assignment highlighted the importance of Artificial Intelligence in modern network management through the discussion of AIOps and other AI-based technologies. The project also promoted the use of indigenous South African languages through the use of subtitling technologies in video presentations.
- Overall, the assignment provided practical experience in network design, system administration, security management, and teamwork while improving our understanding of enterprise network environments and modern IT infrastructure solutions.